Chapter 1: What Is Machine Learning?

Machine learning (ML) is a field of artificial intelligence in which computers learn patterns from data instead of being explicitly programmed with rules. A model is trained on examples and then used to make predictions on new, unseen data.

There are three main types of learning. In supervised learning the training data has labels, and the model learns to predict them; regression predicts a number and classification predicts a category. In unsupervised learning there are no labels and the model finds structure on its own, for example clustering and dimensionality reduction. In reinforcement learning an agent learns by interacting with an environment and receiving rewards or penalties.

A typical ML workflow has these steps: collect data, clean and prepare it, choose features, select a model, train it, evaluate it and finally deploy and monitor it. Data quality usually matters more than the choice of algorithm.

Chapter 2: Linear Regression and Gradient Descent

Linear regression is a supervised algorithm that predicts a continuous value by fitting a straight line to the data. With one feature the model is y = w * x + b, where w is the weight (slope) and b is the bias (intercept). With many features, each feature has its own weight.

The loss function measures how wrong the predictions are. The most common loss for regression is Mean Squared Error (MSE), which is the average of the squared differences between the predicted and the actual values. Training means finding the weights that minimise this loss.

Gradient descent is an optimisation algorithm that does this step by step. At each step it computes the gradient of the loss with respect to each weight and moves the weight a small step in the opposite direction. The size of the step is controlled by the learning rate. If the learning rate is too large the loss may bounce around or diverge, and if it is too small the training becomes very slow.

Chapter 3: Classification and Evaluation Metrics

Classification predicts a discrete label. Logistic regression, despite its name, is a classification algorithm; it passes a linear combination of features through the sigmoid function to produce a probability between 0 and 1. A decision tree splits the data using a sequence of questions about the features. The k-nearest neighbours (k-NN) algorithm labels a new point by the majority label of its k closest training points.

Accuracy is the fraction of predictions that are correct, but it can be misleading when classes are imbalanced. Precision is the fraction of predicted positives that are truly positive: TP / (TP + FP). Recall is the fraction of actual positives that the model finds: TP / (TP + FN). The F1 score is the harmonic mean of precision and recall.

A confusion matrix shows the counts of true positives, false positives, true negatives and false negatives. For a disease screening test, recall is usually more important because missing a sick patient is costly; for a spam filter, precision matters because wrongly blocking a real email is costly.

Chapter 4: Overfitting, Underfitting and Validation

Overfitting happens when a model learns the training data too closely, including its noise, so it performs well on training data but badly on new data. Underfitting happens when the model is too simple to capture the pattern, so it performs badly on both. This trade-off is known as the bias-variance trade-off: simple models have high bias, and complex models have high variance.

To measure generalisation, data is split into a training set, a validation set used to tune hyperparameters, and a test set used only once at the end. K-fold cross-validation splits the data into k parts and trains k times, each time using a different part for validation, which gives a more reliable estimate when data is limited.

Ways to reduce overfitting include collecting more data, using a simpler model, early stopping, dropout in neural networks, and regularisation. L1 regularisation (Lasso) adds the sum of absolute weights to the loss and can push some weights to exactly zero, while L2 regularisation (Ridge) adds the sum of squared weights and shrinks weights smoothly.

Chapter 5: Clustering and Dimensionality Reduction

Clustering groups similar data points without using labels. The k-means algorithm works in a loop: choose k initial centroids, assign every point to its nearest centroid, move each centroid to the mean of its assigned points, and repeat until the assignments stop changing. The value of k must be chosen by the user; the elbow method plots the within-cluster error against k and looks for the bend in the curve.

K-means is sensitive to the starting centroids and to the scale of the features, so features are usually standardised first. Hierarchical clustering builds a tree of clusters (a dendrogram) and does not need k in advance. DBSCAN finds clusters of dense regions and can detect outliers as noise.

Principal Component Analysis (PCA) is a dimensionality reduction technique. It finds new axes, called principal components, that capture the largest amount of variance in the data, and projects the data onto the first few of them. PCA is used to visualise high-dimensional data and to speed up training.

Chapter 6: Neural Networks

A neural network is made of layers of connected units called neurons. Each neuron computes a weighted sum of its inputs, adds a bias and passes the result through an activation function. A network has an input layer, one or more hidden layers and an output layer. A network with many hidden layers is called a deep neural network.

Activation functions add non-linearity, without which the whole network would behave like a single linear model. Popular choices are ReLU, which outputs max(0, x), the sigmoid, which squashes values to between 0 and 1, and softmax, which turns the final layer into class probabilities.

Training uses backpropagation, which applies the chain rule to compute the gradient of the loss with respect to every weight, and then gradient descent updates the weights. One epoch is one full pass through the training data, and the data is usually processed in small groups called mini-batches. Convolutional neural networks (CNNs) are designed for images, and recurrent networks and transformers are designed for sequences such as text.